

Zadanie 28

Oblicz granicę $\lim_{n \rightarrow \infty} \sqrt{\frac{(3^{n-1}+2)(3^{n-1}-2)}{3^{2n}+5}}$.

① Obliczamy granicę.

$$\lim_{n \rightarrow \infty} \sqrt{\frac{(3^{n-1}+2)(3^{n-1}-2)}{3^{2n}+5}} = \lim_{n \rightarrow \infty} \sqrt{\frac{9^{n-1}-4}{9^n+5}} = \lim_{n \rightarrow \infty} \sqrt{\frac{9^n(\frac{1}{9} - (\frac{4}{9^n}))}{9^n(1 + (\frac{5}{9^n}))}} =$$

$$= \lim_{n \rightarrow \infty} \sqrt{\frac{\frac{1}{9}}{1}} = \frac{1}{3}$$

$$\text{Odp.: } \lim_{n \rightarrow \infty} \sqrt{\frac{(3^{n-1}+2)(3^{n-1}-2)}{3^{2n}+5}} = \frac{1}{3}.$$